

EXPERIMENT 27

By Trishil Chauhan

Aim

To write a menu-driven C program using a **switch** statement and nested loops to print five pyramid/diamond patterns. Each pattern is implemented in a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu containing five pyramid/diamond pattern choices.
3. Read the user's choice.
4. Use a switch statement to call the corresponding parameterless void function.
5. In the selected function, input the number of rows n.
6. Use nested loops to print leading spaces and the required stars or numbers.
7. For the diamond, print an upper pyramid followed by a lower inverted half.
8. If the choice is outside 1-5, print an invalid-choice message.
9. Stop.

C Program

```
#include <stdio.h>

void fullPyramid(void);
void invertedPyramid(void);
void diamondPattern(void);
void palindromicPyramid(void);
void repeatedNumberPyramid(void);

int main(void) {
    int choice;

    printf("PYRAMID / DIAMOND PATTERNS MENU\n");
    printf("1. Full pyramid\n");
    printf("2. Inverted pyramid\n");
    printf("3. Diamond pattern\n");
    printf("4. Palindromic number pyramid\n");
    printf("5. Repeated-number pyramid\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: fullPyramid(); break;
        case 2: invertedPyramid(); break;
        case 3: diamondPattern(); break;
        case 4: palindromicPyramid(); break;
        case 5: repeatedNumberPyramid(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

void fullPyramid(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}

void invertedPyramid(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}
```

```

    }
}

void diamondPattern(void) {
    int n, i, j;
    printf("Enter rows for upper half: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
    for (i = n-1; i >= 1; i--) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}

void palindromicPyramid(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        for (j = i-1; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

void repeatedNumberPyramid(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("%d ", i);
        printf("\n");
    }
}

```

Test-Case Table

Test	Choice	n	Expected result
1	1	4	Full pyramid with 1, 3, 5, 7 stars
2	2	4	Inverted pyramid with 7, 5, 3, 1 stars
3	3	4	Diamond of 7 rows; middle row has 7 stars
4	4	4	Palindromic rows ending with 1 2 3 4 3 2 1
5	5	4	Repeated rows: 1; 2 2 2; 3 x 5; 4 x 7
6	8	-	Invalid choice!

Sample Outputs (n = 4)

1. Full pyramid

```

*
***
*****
*****

```

2. Inverted pyramid

```

*****
*****
***
*

```

3. Diamond pattern

```
*  
***  
*****  
*****  
*****  
***  
*
```

4. Palindromic number pyramid

```
1  
1 2 1  
1 2 3 2 1  
1 2 3 4 3 2 1
```

5. Repeated-number pyramid

```
1  
2 2 2  
3 3 3 3 3  
4 4 4 4 4 4 4
```

Result

The program successfully displays all five pyramid/diamond patterns using a switch statement, separate parameterless void functions, and nested loops. The prepared test cases verify every menu option and an invalid menu choice.